Indices, Surds and Polynomials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

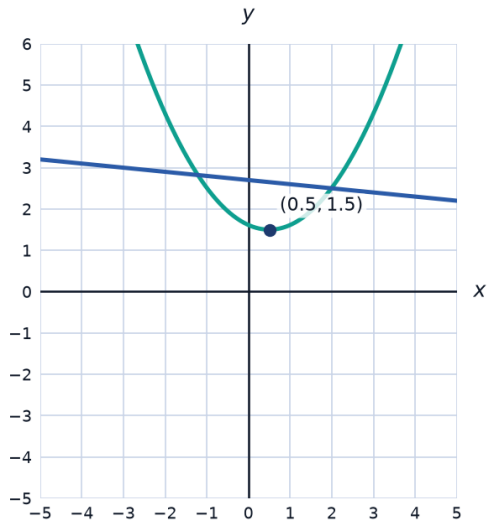
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p+q\sqrt{5}$, where p and q are rational numbers to be found.

2 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

[About A2 Level · Practice 2] If the governing relation for this system is $g(t)=4t+4$, evaluate $g(6)$ and find the inverse input t when $g(t)=32$.



3 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

4 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

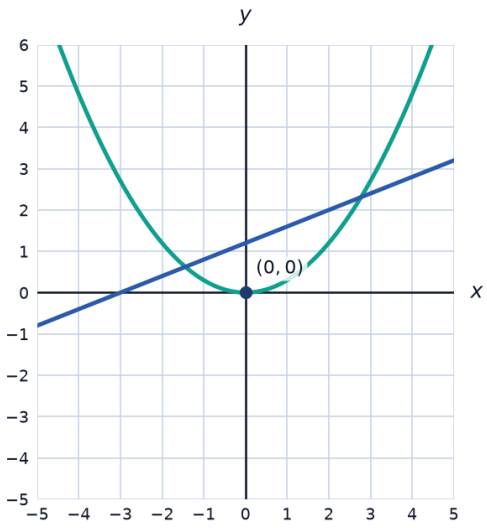
5 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

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6 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

[Differentiation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



7 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

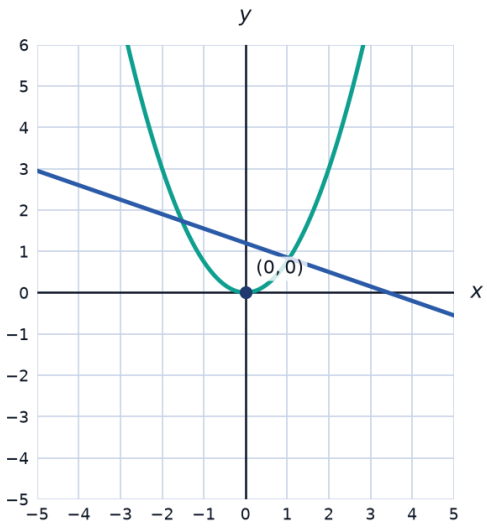
9 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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10 **[Intermediate]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 **[Intermediate]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 **[Intermediate]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 **[Intermediate]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 **[Intermediate]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

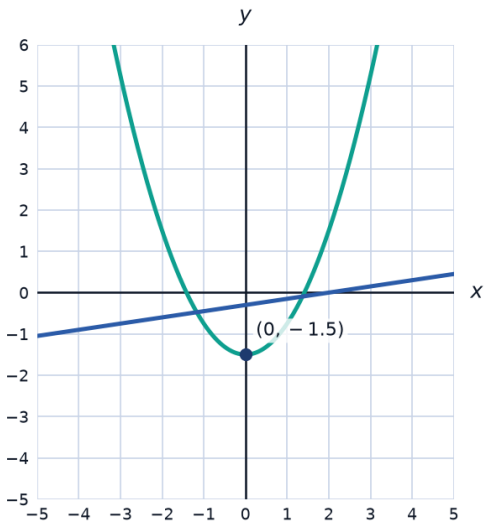
15 **[Challenging]** • **Target: Indices, Surds & Polynomials Practice Exercises** • **Cross-Lesson: 4.01 Logarithmic and Exponential Functions**

[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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16 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

[Algebra & Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



17 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

[Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

[Mensuration · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

19 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

[Number · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

[Sets & Venn Diagrams · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

a. $-\frac{1}{2}$

b. $\frac{1}{2}$

c. $-\frac{9}{2}$

d. 0

22 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

When the polynomial $P(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x - 3$, the remainder is 10. What is the value of the constant a?

a. -3

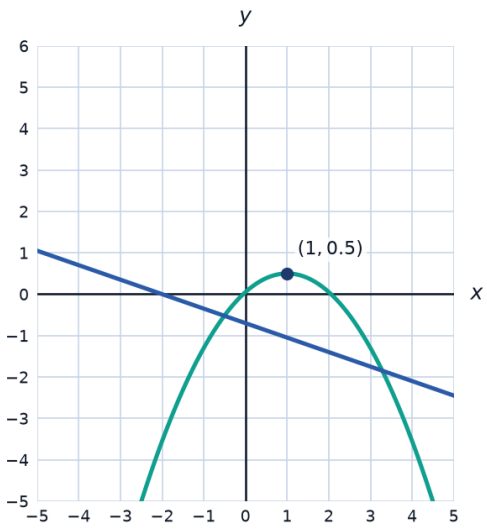
b. 3

c. -9

d. -33

23 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The polynomial $p(x) = 2x^3 + ax^2 - 5x + 6$ is divisible by $x - 2$. Find the value of the constant a.



a. -3

b. 3

c. 0

d. -6

24 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

In the expansion of $(1 + \frac{x}{2})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

a. 9

b. 11

c. 8

d. 5

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25 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 4)$. What is the value of the constant a?

a. -10.00

b. 7.00

c. -7.00

d. -5.00

26 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The function f is defined for all real numbers x by $f(x) = 2 - 3\sin(4x)$. What are the amplitude and the period (in radians) of f?

a. $\text{Amplitude} = 3, \quad \text{Period} = \frac{\pi}{2}$

b. $\text{Amplitude} = -3, \quad \text{Period} = \frac{\pi}{2}$

c. $\text{Amplitude} = 3, \quad \text{Period} = 8\pi$

d. $\text{Amplitude} = 5, \quad \text{Period} = \frac{\pi}{4}$

27 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

a. 14 cm

b. $2\sqrt{19}$ cm

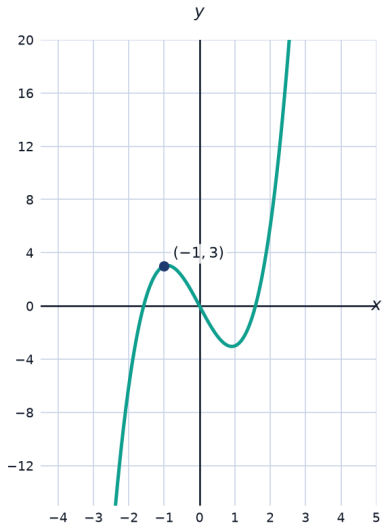
c. $2\sqrt{34}$ cm

d. 8 cm

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28 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.



a. 57 units per second

b. 19 units per second

c. $\frac{19}{3}$ units per second

d. 21 units per second

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The variables x and y are connected by the equation $y = 2x^3 - 5x$. Given that x increases at a constant rate of 4 units s^{-1}, find the rate of change of y at the instant when $x = 2$.

a. 76 units s^{-1}

b. 19 units s^{-1}

c. 28 units s^{-1}

d. 24 units s^{-1}

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30 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

Find the real solution to the equation $\log_3 x + \log_3 (x - 8) = 2$.

a. $x = 9$

b. $x = 9$ and $x = -1$

c. $x = 8.5$

d. $x = 5$

31 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

A curve has equation $y = \frac{kx}{2x - 1}$, where $x \neq \frac{1}{2}$ and k is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that x is increasing at a rate of 0.6 units per second, find the rate of change of y when $x = 2$.

a. -0.2 units per second

b. 0.2 units per second

c. -1.8 units per second

d. -0.6 units per second

32 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

At 14:00, a coastguard radar station detects two vessels, A and B. Relative to the radar station, vessel A has position vector $(-4\mathbf{i} + 6\mathbf{j})$ km and moves with a constant velocity of $(8\mathbf{i} + 5\mathbf{j})$ km h^{-1}. Vessel B has position vector $(20\mathbf{i} - 4\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 10\mathbf{j})$ km h^{-1}, where u is a constant. Given that vessel B intercepts vessel A, find the value of u.

a. $u = -4$

b. $u = 4$

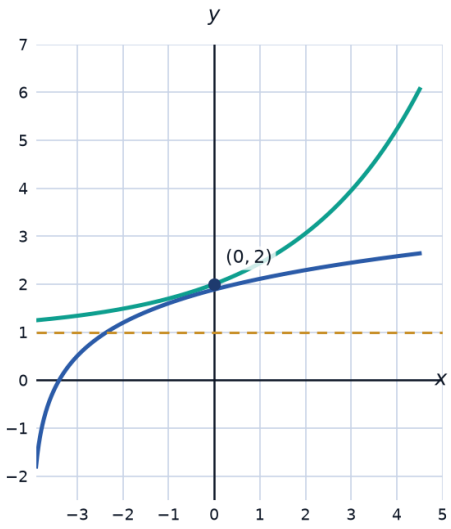
c. $u = -2$

d. $u = -8$

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33 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The variables x and y satisfy the relationship $y = Ab^x$, where A and b are positive constants. The graph of $\ln y$ against x is a straight line passing through the points $(0, \ln 3)$ and $(4, \ln 48)$. What is the value of b?



a. 2

b. 4

c. 16

d. $\sqrt{2}$

34 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The position vectors of points A and B relative to an origin O are given by $\vec{OA} = 5\mathbf{i} - 2\mathbf{j}$ and $\vec{OB} = -\mathbf{i} + 6\mathbf{j}$. Find the magnitude of $\vec{AB}$.

a. $4\sqrt{2}$

b. 10

c. $2\sqrt{13}$

d. 14

35 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the logarithmic equation $\log_2(5x + 6) - \log_2(x) = 3$.

a. $x = 2$

b. $x = 6$

c. $x = \frac{1}{2}$

d. $x = -3$

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36 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The position vectors of points A and B relative to an origin O are $\vec{OA} = 2\mathbf{i} - 3\mathbf{j}$ and $\vec{OB} = 7\mathbf{i} + 9\mathbf{j}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$

b. $\frac{1}{13}(-5\mathbf{i} - 12\mathbf{j})$

c. $\frac{1}{\sqrt{61}}(5\mathbf{i} + 6\mathbf{j})$

d. $\frac{1}{15}(9\mathbf{i} + 12\mathbf{j})$

37 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

Find the value of the integer $n \geq 3$ that satisfies the equation ${}^{n}\mathrm{P}_{3} = 10 \times {}^{n}\mathrm{C}_{2}$

a. $n = 7$

b. $n = 12$

c. $n = 5$

d. $n = 8$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Given that ${}^{n}P_{2} = n3$, where n is an integer such that $n \geq 3$, find the value of n.

a. 8

b. 5

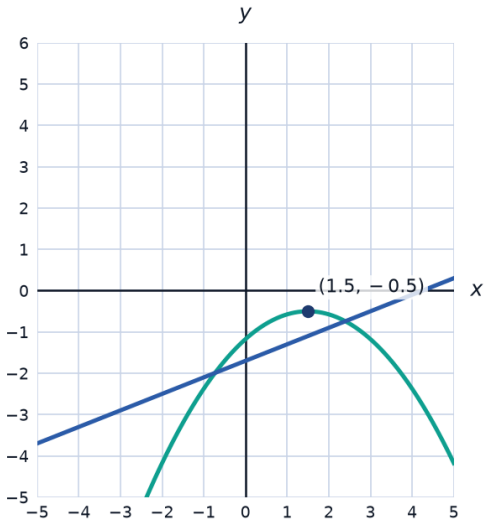
c. 3

d. 6

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39 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 4$. When $t = 1$, the velocity of the particle is 5 m s^{-1}. Find the velocity of the particle when $t = 3$.



a. 15 m s^{-1}

b. 19 m s^{-1}

c. 21 m s^{-1}

d. 27 m s^{-1}

40 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The position vectors of points A and B relative to a fixed origin O are given by $\vec{OA} = -2\mathbf{i} + 5\mathbf{j}$ and $\vec{OB} = 4\mathbf{i} - 3\mathbf{j}$. Find the magnitude of the vector $\vec{AB}$.

a. 10

b. $2\sqrt{2}$

c. 14

d. $2\sqrt{5}$

41 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A sector of a circle of radius 12 cm has a central angle of $\frac{\pi}{3}$ radians. Find the perimeter of the sector, giving your answer in exact form.

a. $(24 + 4\pi)$ cm

b. $(12 + 4\pi)$ cm

c. 4π cm

d. $(24 + 24\pi)$ cm

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42 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A sector of a circle of radius 12 cm has an arc length of 18 cm. Find the area of the sector.

a. $108\ \text{cm}^2$

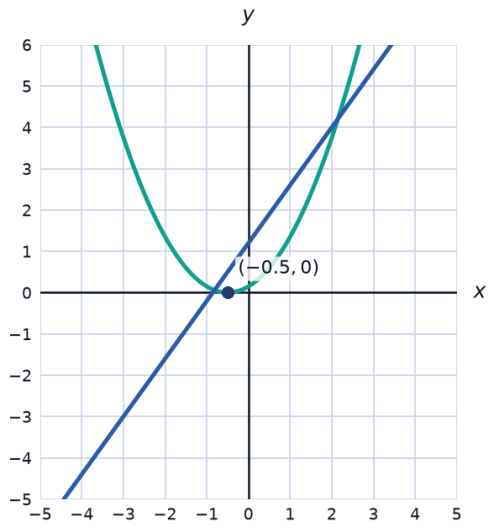
b. $216\ \text{cm}^2$

c. $54\ \text{cm}^2$

d. $144\ \text{cm}^2$

43 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Given that ${}^{n}\text{P}_3 = 8\,{}^{n}\text{C}_2$, find the value of the integer n, where $n \geq 3$.



a. 6

b. 8

c. 10

d. 12

44 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

A particle moves in a straight line with acceleration $a(t) = 6t - 8\ \text{m/s}^2$. If it starts from rest at the origin (at $t = 0$), at what time $t > 0$ does the particle momentarily come to rest again?

a. $t = 4$ s

b. $t = \frac{4}{3}$ s

c. $t = \frac{8}{3}$ s

d. $t = 8$ s

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45 **[Challenging]** • **Target: Indices, Surds & Polynomials Practice Exercises** • **Cross-Lesson: 7.01 Differentiation and Rates of Change**

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 3$ cm?

a. 18π cm^3/s

b. 36π cm^3/s

c. 9π cm^3/s

d. $\frac{18}{2}\pi$ cm^3/s

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